SMARTBRIDGE

SKILL WALLET
A SmartBridge Product

Project Title	Rising Waters: Machine Learning Based Flood Prediction System		
Developer	Meghana Tej Boyina	Team ID	[Team ID]
Date	27 June	Phase / Document	Project Planning Phase

11. Project Planning

As an individually executed project, planning was organized as a sequential personal milestone plan rather than multi-person sprint ceremonies. Each milestone below was treated as a hard prerequisite for the next — for example, the frontend could not be meaningfully wired up until the /predict route contract (which form fields it expects, what it returns) was stable, and that contract could not be finalized until the training pipeline had settled on its final feature set.

Milestone	Description	Status
Data Preparation	Load flood_prediction.csv, rename columns to match app field names, handle missing values via median imputation	Complete
Exploratory Analysis	Dataset inspection and experimentation in notebook.ipynb	Complete
Model Training & Comparison	Train Decision Tree, Random Forest, KNN, and XGBoost; compare accuracy	Complete
Model Selection & Serialization	Select best-performing model; save model + scaler via Joblib	Complete
Backend Development	Build Flask app.py with / and /predict routes	Complete
Frontend Development	Build responsive glass morphism UI in templates/index.html	Complete
Integration & Testing	Connect frontend form to backend prediction pipeline; validate severity/insight logic	Complete
Documentation	README.md and project submission documentation	In Progress

Documentation was intentionally scheduled as the final milestone rather than an ongoing background task, on the basis that writing accurate documentation is far easier once the code, feature set, and UI behaviour have stopped changing — documenting a moving target tends to produce material that goes stale before submission.